# Wrapping up the soil-moisture foundation-model study

What was done, where it sits in the literature, the paper's arguments, cheap additions, and the next paper

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## 1 What was done

**Data.** 990 stations from ISMN (Dorigo et al., 2021), ICOS and AmeriFlux with at least three years of daily data, 2016–2025. Hourly in-situ soil moisture (SM) was QC-filtered (G/D* flags), averaged to daily values and binned to three depths: 0–10, 10–30 and 30–100 cm (885, 682 and 551 stations; 5.14 M station-days). Splits by station: 573 train, 89 validation, 224 out-of-sample (OOS) stations. Splits by time: 2016–2022 train, 2023–2025 out-of-time (OOT). OOST is the joint case (unseen station, unseen years). Normalisation statistics were fit on training stations and years only.

**Inputs.** Per station, a 2.24 km tile:

- frozen TerraMind (Jakubik et al., 2025) layer-12 tokens ($14 \times 14$ at 160 m) for Sentinel-2 L2A (272 k cloud-screened scenes), Sentinel-1 RTC (497 k scenes), DEM and LULC;
- ERA5-Land, 18 variables, 365-day window;
- SIF (last 50 values), GRACE TWSA (12 months), gridded soil properties;
- a light 20 m CNN path on the latest S2/S1 scene (added in §48).

**Model.** A single 3-layer transformer (26.7 M trainable parameters) over 1,038 tokens: three depth query tokens, static tokens, satellite history tokens and daily driver tokens. Huber loss at the station pixel; ERA5 day-dropout 0.15 and modality dropout (ERA5 0.5, satellite 0.3, fine path 0.2). Selected checkpoint: epoch 10, run `baseline_selected_20261005` (runbook §59/§66).

**Table 1:** Baseline skill, station-mean ubRMSE ($m^3\,m^{-3}$) and Pearson $R$ per depth. Source: `eval_output/baseline_selected_20261005/metrics_summary.csv`.

| | | ubRMSE | | | $R$ | | |
|---|---|---|---|---|---|---|---|
| Split | $n$ (0–10) | 0–10 | 10–30 | 30–100 | 0–10 | 10–30 | 30–100 |
| Validation (new stations) | 89 | .0529 | .0476 | .0470 | .753 | .748 | .711 |
| OOS (new stations) | 221 | .0523 | .0483 | .0471 | .722 | .708 | .634 |
| OOT (seen stations, 2023–25) | 362 | .0490 | .0433 | .0437 | .763 | .768 | .720 |
| OOST (new stations, 2023–25) | 114 | .0535 | .0474 | .0495 | .733 | .741 | .688 |

**Table 2:** Input reliance on 221 OOS stations: ubRMSE increase when one input group is swapped with another station's (cross) or with another date at the same station (within). Source: `.../baseline_selected_20261005_ablation/ablation_summary.csv`.

| Input swapped | 0–10 | 10–30 | 30–100 |
|---|---|---|---|
| All satellite tokens, cross | +36.8 % | +37.1 % | +38.4 % |
| ERA5-Land, within station | +39.0 % | +30.4 % | +13.5 % |
| ERA5-Land, cross | +30.3 % | +24.6 % | +15.1 % |
| Soil properties, cross | +7.6 % | +17.6 % | +25.7 % |
| DEM, cross | +12.6 % | +9.7 % | +18.4 % |
| LULC, cross | +6.1 % | +6.6 % | +12.7 % |
| S1 history, cross | +4.8 % | +7.7 % | +14.4 % |
| S2 history, cross | +4.9 % | +6.8 % | +8.7 % |
| S1 history, within station | 0.0 % | +1.0 % | 0.0 % |
| SIF, cross | +1.3 % | +1.6 % | +1.8 % |
| 20 m fine path, cross | +0.5 % | +1.3 % | +0.1 % |
| TWSA, cross | 0.0 % | 0.0 % | +0.1 % |

**Table 3:** Negative and null results already measured.

| Test | Result |
|---|---|
| LST as auxiliary target (§53/§56) | Hurts SM: best val ubRMSE 0.0537 with LST vs 0.0487 without; $R$(0–10) .694 vs .761. |
| 20 m fine path (§56.2, §67) | Unused for SM: val 0.0487 intact, 0.0489 zeroed, 0.0496 shuffled. |
| Landsat LST (§29) | Tile pattern is static (month-to-annual $r = +0.967$); pooled $r = +0.167$ but within-station $-0.077$ (Simpson's paradox). |
| ECOSTRESS DTR (G0) | DTR is the daytime pattern ($r = +0.816$); day vs night $+0.059$. Thermal arm closed. |
| Effective resolution (§70) | $L_{\text{eff}} = 2240$ m at 0–10 cm; within-tile contrast $\sim 4\times$ below observed. |
| Patchwise model (§35, §56) | OOT beat validation (0.0434 vs 0.0507), best epoch 2: site memorisation. Tile-pair sign test 0.50. |
| Depth, wd, regularisation (§60–§65) | 1/2/3/6 layers: 0.0510/0.0494/0.0488/0.0500; wd 0.3: 0.0493; static dropout 0.0513; spectral indices 0.0511. None beats the 3-layer baseline. |
| TerraMind layer (§27b) | kNN skill for station-mean SM: L12 9.8 %, L9 6.9 %, L6 3.6 %, L3 4.0 %; tabular covariates 23.4 %. |

## 2 Literature and the gap

**Geospatial foundation models (GFMs).** Prithvi-EO-2.0 (Szwarcman et al., 2024), SatMAE (Cong et al., 2022), Galileo (Tseng et al., 2025), Copernicus-FM (Wang et al., 2025), DOFA (Xiong et al., 2024), AnySat (Astruc et al., 2024), Presto (Tseng et al., 2023) and TerraMind (Jakubik et al., 2025) are pretrained self-supervised encoders. AlphaEarth (Brown et al., 2025) and TESSERA (Feng et al., 2025) release pre-computed embeddings. Benchmarks (Lacoste et al., 2023; Marsocci et al., 2024; Wang et al., 2025) are dominated by classification and segmentation; PANGAEA finds GFMs do not consistently beat supervised UNets, and Xu et al. (2025) show tuned supervised baselines match specialised foundation models. Rolf et al. (2024) argue for EO-specific, spatially aware evaluation, and SatCLIP (Klemmer et al., 2025) shows that location alone predicts many regression targets.

**GFM embeddings for continuous variables.** The closest SM study (Kontogiorgakis et al., 2026, preprint; 113 European stations) finds Prithvi embeddings add nothing over hand-crafted S1/S2/ERA5 features ($R^2$ 0.515 vs 0.514). An IBM abstract fine-tuning Prithvi on TxSON (Chavda and Das, 2024) reports no fine-tuning numbers. AlphaEarth embeddings are highly stable between years ($r = 0.963$; Rahman, 2026, preprint), i.e. largely static. A buffered spatial validation halves the embedding gain (Ratsakatika et al., 2026), cross-country crop yield gives no FM advantage (Adjei, 2026), and frozen encoders underperform

for biomass while precomputed embeddings do better (Sialelli et al., 2026). A 2026 review (Yu et al., 2026) notes that pretraining corpora lack microwave and thermal data and that embeddings are mostly static while ecohydrology needs dynamics. Fang et al. (2026) (Section 3) is the clearest positive case.

**ML soil moisture from in-situ networks.** SoMo.ml (O and Orth, 2021) trains an LSTM on ISMN to 50 cm at 0.25°; GLASS SM (Zhang et al., 2023) and GSSM1km (Han et al., 2023) produce 1 km surface products; Zhang et al. (2023) is one of few to report random, site-independent and year-independent validation side by side. Deep-learning comparisons (Fang et al., 2017; Wang et al., 2024) use hand-crafted inputs. S1/SMAP downscaling (Das et al., 2019; Bauer-Marschallinger et al., 2019) is surface-only and degrades under dense vegetation.

**Validation.** Random CV overstates skill for spatial prediction (Ploton et al., 2020; Meyer and Pebesma, 2022); the area of applicability (AOA) of Meyer and Pebesma (2021) has been applied to in-situ SM upscaling (Yu et al., 2025).

**Gaps this study fills.**

G1. No test of a multimodal any-to-any GFM (TerraMind) for soil moisture; existing tests use Prithvi only.

G2. No GFM soil-moisture study below the surface; we predict 10–30 and 30–100 cm.

G3. Existing GFM soil-moisture studies use 40–113 stations in one region; we use ∼990 stations across climates.

G4. No study separates the embedding as a site descriptor from the embedding as a dynamic signal.

G5. New-station, new-year and joint generalisation are rarely reported together.

G6. Whether the embedding gain survives a station holdout is untested for soil moisture.

G7. No fusion of GFM embeddings with reanalysis, SIF and GRACE in one temporal model.

G8. GFM benchmarks lack a regression task on a dynamic variable at point sites.

G9. Frozen GFM, precomputed embeddings and raw features are not compared on the same SM task.

G10. No AOA or embedding-distance uncertainty for GFM-based SM at unseen stations.

Gaps G1–G5 and G7 are covered by work already done. G6, G9 and G10 need the additions in Section 5. G8 is the next paper.

# 3 Fang et al. (2026) and what we take from it

**What they do.** AlphaEarth annual 64-d embeddings at 10 m are masked to corn or soybean (CDL) and averaged per county-year (6,350 records, 992 Corn Belt counties, 2017–2024). An RBF-SVR is the only model. Validation is leave-one-year-out plus a roll-forward forecast (train $\leq t-1$). Corn: mean $R^2 = 0.825$, RMSE 14.8 bu/ac; soybean $R^2 = 0.814$. Baselines: training-years county mean ($R^2 \approx 0.69$) and NDVI+PRISM+GDD (0.687 corn, 0.732 soy). Diagnostics: Moran's $I$ of relative errors 0.32–0.49 every year (spatially clustered residuals, read as missing slow drivers); PCA axes explain $<3$ % of error variance (variance $\neq$ task relevance); test-fold permutation importance on the 64 dimensions; a label-subsampling curve.

**Table 4:** Fang et al. (2026) vs this study.

| | Fang et al. (2026) | This study |
|---|---|---|
| Target | County yield, once per year | Daily SM at a point, 3 depths |
| Embedding | AlphaEarth, annual, 64-d, precomputed | TerraMind L12 tokens per scene, frozen |
| Other inputs | None | ERA5-Land, SIF, TWSA, soil |
| Model | RBF-SVR | 3-layer transformer |
| Spatial holdout | None (every county in every training fold) | 89 val + 224 OOS stations |
| Temporal holdout | LOYO, roll-forward | OOT 2023–25, OOST |
| Naive baseline | County mean, NDVI+weather | Not yet (Section 5, E1) |
| Residual diagnostics | Moran's $I$, LISA | Not yet (E3) |

**How it relates.** Both studies ask whether GFM embeddings are analysis-ready features. Their setting cannot separate site identity from signal: every county is in training, the county mean alone gives $R^2 \approx 0.69$, and an annual embedding has no within-season dynamics. Our result fills exactly that hole: under a station holdout the embeddings carry place, and the time-varying skill comes from the drivers (Table 2). We cite Fang et al. as the positive case and position our paper as the test it lacks.

**What we borrow.** (i) A per-station climatology baseline, the analogue of their county mean. (ii) Moran's $I$ / LISA on OOS station residuals. (iii) A label-subsampling curve. (iv) AlphaEarth embeddings as a drop-in static token, which also gives a GFM-vs-GFM comparison (G9).

# 4 Paper storyline and arguments

**Working title.** *Frozen foundation-model embeddings describe the site, weather drives the dynamics: daily root-zone soil moisture at 990 stations.*

**One-paragraph story.** Frozen TerraMind embeddings fused with reanalysis and gravimetric drivers predict daily soil moisture to 1 m at unseen stations with ubRMSE 0.047–0.052 $m^3\ m^{-3}$, close to stations seen in training. Swap tests show the embeddings act as a static landscape descriptor that conditions the weather-to-soil-moisture response; they do not carry the temporal signal. Reliance shifts from weather to soil and terrain with depth. Finer resolution and thermal inputs do not help. This agrees with recent reports that GFM embeddings are static, and adds the first station-holdout, multi-depth evidence for soil moisture.

**Arguments, each with its number.**

A1. **Transfer to new sites is close to seen sites.** OOS ubRMSE .052/.048/.047 vs validation .053/.048/.047 and OOT .049/.043/.044 (Table 1).

A2. **Embeddings carry place, not process.** Swapping all satellite tokens across stations costs +37–38 %; swapping S1 history within a station costs 0–1 %. L12 tokens are nearly static across seasons (§72.2). Consistent with Rahman (2026); Yu et al. (2026).

A3. **Dynamics come from the drivers.** ERA5 swap within station +39 % at 0–10 cm, falling to +13.5 % at 30–100 cm.

A4. **Reliance inverts with depth.** Soil properties go from +7.6 % to +25.7 %, DEM to +18.4 % at 30–100 cm.

A5. **More resolution and thermal do not help.** Fine path 0.1–1.3 %; LST target worsens SM by 10 %; effective resolution 2.2 km, the tile size.

A6. **Validation design decides the verdict.** The patchwise model looked good until OOT beat validation, a memorisation signature. We report new-station, new-year and joint splits together (Ploton et al., 2020; Meyer and Pebesma, 2022).

**Wording rule.** The swap tests measure *reliance*, not *added value*: a model trained without an input could compensate. Say "the model relies on" until the drivers-only run (E2) exists.

**Open bookkeeping.** OOS counts differ: 224 in the split, 222 evaluated, 221 at 0–10 cm. The ablation file's reference ubRMSE (.0500/.0479/.0476) uses a different aggregation from Table 1, so quote only relative changes from it. Fix both before submission.

## 5 Cheap additions for this paper

Ranked by cost. Eval-only items reuse the existing predictions.

| | Addition | Cost | Why |
|---|---|---|---|
| E1 | Naive baselines: station climatology (OOT only), persistence, ERA5-only gradient boosting | CPU, eval | Reviewers will ask; we have none. Mirrors Fang's county mean. |
| E2 | Drivers-only model (no satellite tokens), 10 epochs | 1 training run | Turns "reliance" into "added value". The most important addition. |
| E3 | Moran's $I$ / LISA on OOS residuals | CPU, eval | Shows whether errors cluster in space (missing slow drivers). |
| E4 | AOA: embedding distance to training stations vs OOS error | CPU, eval | Uncertainty for new sites (G10). |
| E5 | Group Shapley (§73, already planned) | GPU, eval | Clean attribution per input group. |
| E6 | Label-subsampling: train on 25 % and 50 % of stations | 2 runs | How much data the embeddings save. Optional. |
| E7 | AlphaEarth annual embedding as a static token | GEE pull + 1 run | GFM-vs-GFM (G9). Optional. |

Minimum for submission: E1, E2, E3. E4–E7 strengthen it but are not required.

## 6 The next paper

N1. **A soil-moisture benchmark for GFMs.** ISMN stations, fixed new-station / new-year / joint splits, three depths. Encoders: TerraMind, Prithvi, AlphaEarth, TESSERA, Galileo, raw bands. This fills G8 and reuses the pipeline that already exists.

N2. **Is "static" a pretraining limit?** Fine-tune the encoder, or use time-resolved embeddings (Presto/Galileo-style pixel time series), and test whether the within-station swap effect rises above zero.

N3. **Wall-to-wall maps with uncertainty.** The Overijssel run (§68) with TWENTE validation and an AOA mask.

N4. **Management signals.** Irrigation at Mead (§69): the irrigation signal appears only in dry years, and S2 does not see it.

## References

K. Adjei. Do foundation model embeddings improve cross-country crop yield generalisation?, 2026. Preprint, arXiv:2605.08113.

Guillaume Astruc, Nicolas Gonthier, Clement Mallet, and Loic Landrieu. AnySat: One earth observation model for many resolutions, scales, and modalities, 2024. arXiv:2412.14123.

Bernhard Bauer-Marschallinger et al. Toward global soil moisture monitoring with Sentinel-1: Harnessing assets and overcoming obstacles. *IEEE Transactions on Geoscience and Remote Sensing*, 57:520–539, 2019.

Christopher F. Brown, Michal R. Kazmierski, Valerie J. Pasquarella, et al. AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data, 2025. arXiv:2507.22291.

J. Chavda and K. Das. Soil moisture estimation with geospatial foundation models and satellite imagery. AGU Fall Meeting abstract, IBM Research, 2024.

Yezhen Cong, Samar Khanna, Chenlin Meng, et al. SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery. In *Advances in Neural Information Processing Systems*, volume 35, pages 197–211, 2022.

Narendra N. Das et al. The SMAP and Copernicus Sentinel 1A/B microwave active-passive high resolution surface soil moisture product. *Remote Sensing of Environment*, 233:111380, 2019. doi: 10.1016/j.rse.2019.111380.

Wouter Dorigo et al. The International Soil Moisture Network: serving Earth system science for over a decade. *Hydrology and Earth System Sciences*, 25:5749–5804, 2021. doi: 10.5194/hess-25-5749-2021.

Jichao Fang, Mingda Wu, Zhou Zhang, and Wei Luo. Application of AlphaEarth geofoundation model for high-accuracy crop yield estimation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*, pages 1465–1474, 2026.

Kuai Fang, Chaopeng Shen, Daniel Kifer, and Xiao Yang. Prolongation of SMAP to spatiotemporally seamless coverage of continental U.S. using a deep learning neural network. *Geophysical Research Letters*, 44:11030–11039, 2017.

Zhengpeng Feng, Srinivasan Keshav, et al. TESSERA: Temporal embeddings of surface spectra for earth representation and analysis, 2025. arXiv:2506.20380.

Qianqian Han, Yijian Zeng, et al. Global long term daily 1 km surface soil moisture dataset with physics informed machine learning. *Scientific Data*, 10:101, 2023. doi: 10.1038/s41597-023-02011-7.

Johannes Jakubik et al. TerraMind: Large-scale generative multimodality for earth observation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. arXiv:2504.11171.

Konstantin Klemmer, Esther Rolf, Caleb Robinson, Lester Mackey, and Marc Rußwurm. SatCLIP: Global, general-purpose location embeddings with satellite imagery. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.

I. Kontogiorgakis et al. Comparative assessment of multimodal earth observation data for soil moisture estimation, 2026. Preprint, arXiv:2602.18083.

Alexandre Lacoste et al. GEO-Bench: Toward foundation models for earth monitoring. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks)*, pages 51080–51093, 2023.

Valerio Marsocci et al. PANGAEA: A global and inclusive benchmark for geospatial foundation models, 2024. arXiv:2412.04204.

Hanna Meyer and Edzer Pebesma. Predicting into unknown space? estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution*, 12:1620–1633, 2021.

Hanna Meyer and Edzer Pebesma. Machine learning-based global maps of ecological variables and the challenge of assessing them. *Nature Communications*, 13:2208, 2022. doi: 10.1038/s41467-022-29838-9.

Sungmin O and Rene Orth. Global soil moisture data derived through machine learning trained with in-situ measurements. *Scientific Data*, 8:170, 2021. doi: 10.1038/s41597-021-00964-1.

Pierre Ploton et al. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications*, 11:4540, 2020.

M. Rahman. Physically interpretable AlphaEarth embeddings, 2026. Preprint, arXiv:2602.10354.

A. Ratsakatika, M. Zotta, Srinivasan Keshav, and J. Lines. Geospatial embeddings under buffered spatial validation, 2026. Preprint, arXiv:2609.28194.

Esther Rolf, Konstantin Klemmer, Caleb Robinson, and Hannah Kerner. Position: Mission critical – satellite data is a distinct modality in machine learning. In *Proceedings of the 41st International Conference on Machine Learning (ICML), PMLR 235*, 2024.

Ghjulia Sialelli, Linus Scheibenreif, Jan Dirk Wegner, and Konrad Schindler. Above-ground biomass estimation with geospatial foundation models, 2026. Preprint, arXiv:2608.04792.

Daniela Szwarcman et al. Prithvi-EO-2.0: A versatile multi-temporal foundation model for earth observation applications, 2024. arXiv:2412.02732.

Gabriel Tseng et al. Lightweight, pre-trained transformers for remote sensing timeseries, 2023. arXiv:2304.14065.

Gabriel Tseng et al. Galileo: Learning global and local features of many remote sensing modalities. In *Proceedings of the 42nd International Conference on Machine Learning (ICML), PMLR 267*, 2025.

Y. Wang, L. Shi, et al. A comprehensive study of deep learning for soil moisture prediction. *Hydrology and Earth System Sciences*, 28:917–943, 2024. doi: 10.5194/hess-28-917-2024.

Yi Wang et al. Towards a unified Copernicus foundation model for earth vision, 2025. arXiv:2503.11849.

Zhitong Xiong et al. Neural plasticity-inspired multimodal foundation model for earth observation, 2024. arXiv:2403.15356.

Zongzhe Xu et al. Specialized foundation models struggle to beat supervised baselines. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2411.02796.

X. Yu, B. Malone, J. P. Walker, et al. Spatial soil moisture prediction from in-situ data upscaled to Landsat footprint. *IEEE Transactions on Geoscience and Remote Sensing*, 63, 2025. doi: 10.1109/TGRS.2025.3565818.

X. Yu, J. Peng, X. Lin, T. F. Keenan, and J. Bishop. Earth observation foundation models for terrestrial ecohydrology: From representation learning to process inference, 2026. Preprint, arXiv:2608.15282.

Yufang Zhang, Shunlin Liang, et al. A global long-term daily 1 km surface soil moisture product (glass sm). *Earth System Science Data*, 15:2055–2079, 2023. doi: 10.5194/essd-15-2055-2023.